# Interactive degree-planning simulator: model US-to-ECTS credit transfers and term-calendar overlaps across 20 global universities

Computer
Term-calendar and break-length comparison matrix: 20 global universities mapped by semester dates, holiday duration, and credit load
Computer
Run the test suite locally
Explore the integration overlay files
Add a university to the dataset

Ran 20 tests in the execution environment—all passed—and prepared a University of Washington dataset addition plus tested calendar-overlap logic. The previous ZIPs were unavailable in this runtime, so I reconstructed the earlier 12-test suite from the visible source rather than claiming to rerun the original archive.

The update ZIP, Washington dataset entry, and test output are attached. No remote repository or existing Semester files were changed.

### Test results

| Group | Tests | Result |
|---|---|---|
| Credit conversions and input validation | 9 | Passed |
| Degree baseline and conditional transfer scenarios | 3 | Passed |
| Calendar overlaps, break duration, and invalid dates | 8 | Passed |
| Total | 20 | Passed |

The new calendar tests cover:

- Disjoint periods returning zero overlap.
- A shared endpoint counting as one day.
- A contained nine-day interval.
- Breaks crossing a year boundary.
- Leap-day handling.
- Invalid, reversed, and missing dates.

These tests verify calculation functions—not browser rendering, accessibility, or integration with Semester.

### Overlay file review

Based on the previously generated source—not a fresh inspection of the unavailable ZIP—the integration overlay contains:

| File | Responsibility |
|---|---|
| `index.html` | Matrix and calculator page markup |
| `app.js` | Filtering, sorting, calculator interaction, and dataset loading |
| `core.mjs` | Pure credit-conversion and degree-planning functions |
| `style.css` | Prototype styling |
| `universities.json` | University reference data |

Its proposed destination remains `app/public/credit-planner/`. It does not register an authenticated Semester screen, add navigation, or write official student records.

The update adds a separate `calendar.mjs` module. Its overlap logic is tested, but it is **not** wired into the browser simulator yet.

### University addition

I selected the University of Washington to add a quarter-system example with verified 2026–27 dates. [1]

| Term | Instruction | Exams |
|---|---|---|
| Autumn | Sept. 30–Dec. 11, 2026 | Dec. 12–18 |
| Winter | Jan. 4–Mar. 12, 2027 | Mar. 13–19 |
| Spring | Mar. 29–June 4, 2027 | June 5–11 |
| Summer, full term | June 21–Aug. 20, 2027 | Typically the last class day |

These are Washington's published instruction and examination periods. [1]

| Break | Dates | Inclusive calendar days |
|---|---|---|
| Winter | Dec. 19, 2026–Jan. 3, 2027 | 16 |
| Spring | Mar. 20–28, 2027 | 9 |
| Summer | June 12–20, 2027 | 9 |
| Autumn | Aug. 21–Sept. 28, 2027 | 39 |

The day counts are calculated from the published break dates; they are calendar days, not lost class-meeting days. [1]

Washington's US-semester-to-ECTS ratio and normal credit load remain unverified in the entry. Quarter credits must not be silently treated as semester hours.

### Apply the update locally

Extract the update ZIP, then run its included script against your previously downloaded dataset:

```
python3 add_university.py \
  /path/to/semester-credit-planner/universities.json
```

For the Semester integration copy:

```
python3 add_university.py \
  /path/to/semester/app/public/credit-planner/universities.json
```

The script creates a backup and refuses duplicate Washington entries. Applying it to the original 20-row dataset produces 21 rows; that update has not been applied to your local machine.

Run the reconstructed and extended suite from the update directory:

```
node --test tests/core.test.mjs
```

Full cross-university overlap comparison still requires verified ISO start/end dates for the other institutions. The existing month ranges cannot support precise overlap counts without introducing unsupported assumptions.

⁂

1. https://www.washington.edu/students/reg/2627cal.html